

SWaT Local Data Drift EDA

Train-Test Distribution Drift Summary

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Research Question

Check whether SWaT already exists under `data/` in the current workspace. If it does, run a short EDA focused on KL divergence and other distribution-drift metrics between the train sequence and the test sequence.

Executive Summary

SWaT is already present locally under `data/SWaT/` with three CSV files: `normal.csv`, `attack.csv`, and `merged.csv`.

For the local data that is actually present, the effective split is:

- train sequence: `data/SWaT/normal.csv`
- test sequence: `data/SWaT/attack.csv`

The key caveat is important: the local `attack.csv` is fully labeled `Attack`, not a mixed `normal-plus-attack` test stream. So the measured drift is intentionally very large because the comparison is effectively **normal-only train versus attack-only test**.

To avoid relying on one metric only, the EDA computed:

- histogram-based `KL(train || test)`
- histogram-based `KL(test || train)`
- Jensen-Shannon divergence
- Wasserstein distance
- Kolmogorov-Smirnov statistic
- standardized absolute mean shift

The strongest drift features are concentrated around AIT201, AIT402, PIT501, PIT502, PIT503, AIT501, and AIT504.

Local Data Presence

- `data/SWaT/normal.csv`
- `data/SWaT/attack.csv`

- data/SWaT/merged.csv

Approximate local directory size: 815M.

Data Preparation

The EDA used the local CSV files directly.

- Column names were stripped because some SWaT columns contain leading spaces.
- Timestamp and Normal/Attack were excluded from feature drift calculations.
- The remaining 51 columns were coerced to numeric.
- Parse gaps were forward-filled and backward-filled.

This is broadly consistent with the reference SWaT loader in M2N2, which also loads SWaT from data/SWaT and applies numeric coercion plus forward fill.

Dataset Overview

Train rows	1,387,098
Test rows	54,621
Merged rows	1,441,719
Feature count	51
Test label counts	Attack = 54,621
Merged label counts	Normal = 1,387,098; Attack = 54,621
Attack fraction in merged	0.037886023559375995
Train time start	28/12/2015 10:00:00 AM
Train time end	28/12/2015 9:59:59 AM
Test time start	28/12/2015 10:29:14 AM
Test time end	2/1/2016 1:41:11 PM

Train vs Test Attack Drift

For train_normal_vs_test_attack:

Median KL(train test)	0.9119754221240752
Mean KL(train test)	1.7095001072426963
Median KL(test train)	1.1404711982143674
Mean KL(test train)	1.5535797693816136
Median Jensen-Shannon divergence	0.20457869744797724
Mean Jensen-Shannon divergence	0.18006588977272778
Median Wasserstein distance	0.6047679787452024
Mean Wasserstein distance	20.81701787233951
Median KS statistic	0.5451787978701341

Mean KS statistic	0.40667220913158286
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Important Metric Caveat

The raw mean standardized mean shift is not reliable as a global summary because several actuator channels are constant in train, so division by near-zero train standard deviation explodes numerically.

Near-zero-train-variance features:

- P202
- P204
- P206
- P401
- P404
- P502
- P601
- P603

If those constant-train features are excluded:

- filtered mean standardized mean shift: 3.5015085575606877
- filtered median standardized mean shift: 1.3971969600481424

Strongest Drift Features

Top 5 by Jensen-Shannon divergence

Feature	JS	KL(train test)	KS
AIT201	0.5340048539	10.0911215422	0.7331484312
AIT402	0.4188602422	2.7303052156	0.5813978524
PIT502	0.3957472820	3.2242747465	0.6385688719
AIT501	0.3699596410	2.6227871986	0.6444240137
PIT503	0.3626607116	14.7790194401	0.6305090183

Top 5 by KL(train || test)

Feature	KL(train test)	JS	KS
PIT503	14.7790194401	0.3626607116	0.6305090183
AIT201	10.0911215422	0.5340048539	0.7331484312
PIT501	8.8319918856	0.3498112692	0.6305787789
AIT504	4.1594030789	0.2577998092	0.6823534949
AIT401	4.1422876876	0.0423080941	0.3295489740

Top 5 by KS statistic

Feature	KS	JS	KL(train test)
P201	0.7402253670	0.3509060820	2.6884961379
AIT201	0.7331484312	0.5340048539	10.0911215422
AIT504	0.6823534949	0.2577998092	4.1594030789
MV201	0.6460041581	0.2452556668	0.9415826520
AIT501	0.6444240137	0.3699596410	2.6227871986

Train vs Merged Drift

For `train_normal_vs_merged_all_rows`, drift is much smaller:

Median KL(train merged)	0.0027561185712303214
Mean KL(train merged)	0.006210088341160504
Median Jensen-Shannon divergence	0.0007448783128877291
Mean Jensen-Shannon divergence	0.0019229209169445434
Median KS statistic	0.02065465678018019
Mean KS statistic	0.015407192896102635

This is consistent with the fact that `merged.csv` is dominated by the normal portion, and only about 3.79% of rows are labeled Attack.

Interpretation

- Comparing `normal.csv` against `attack.csv` produces a strong drift signal across many features. This is expected and not surprising.
- The strongest drift is not evenly spread across all channels. A smaller set of sensors and pressure-related channels dominates the divergence signal.
- Comparing `normal.csv` against `merged.csv` produces only mild drift because the merged file is mostly normal data.
- For future benchmark integration, the codebase should treat “attack-only test” and “mixed-timeline test” as two different evaluation semantics.

Artifacts

- Summary JSON: `outputs/eda_swat_drift_2026-06-24/swat_drift_summary.json`
- Per-feature metrics: `outputs/eda_swat_drift_2026-06-24/train_vs_test_attack_feature_metrics.c`
- Per-feature metrics: `outputs/eda_swat_drift_2026-06-24/train_vs_merged_feature_metrics.c`
- Research note: `documents/logs/06-24-2026/research/research-swat-local-data-drift-eda.md`

Reference Loader Path

Reference SWaT loader used for alignment:

- `bsc-thesis-ref-codebases/M2N2-master/data/load_data.py`
- It expects:
 - `data/SWaT/SWaT_Dataset_Normal_v0.csv`
 - `data/SWaT/SWaT_Dataset_Attack_v0.csv`